

Nikki Phach

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Skills

TypeScript, JavaScript, Python, SQL, React, React Router, Redux, HTML/CSS, XState, TailwindCSS, shadcn/ui, Responsive Web Design, FastAPI, Flask, REST APIs, Phaser, Vitest, Unit Testing, Zod, npm, Docker, Git/GitHub, Java, C, Linux/Unix, PyTorch, NumPy, Pandas, scikit-learn, HuggingFace, Jenkins, Machine Learning, Natural Language Processing, Computer Vision, Data Structures/Algorithms, Database Design, Data Processing

Work Experience

Konami Gaming Inc., Software Engineer I

April 2025 - present

- Eliminated ~20 hours of the math team's workload per week by developing an internal tool that automated maximum liability calculations, enabling product managers to directly request and update game liability data.
- Enabled promotional free spin capability for 60+ online casinos by developing a universal feature that integrates with a third-party content provider's API and UI wrapper.
- Ensured compliance with Brazil gaming regulations by introducing a user confirmation flow for automatic spins within gameplay.
- Improved flexibility of RTP (return-to-player) configuration by redesigning the loading flow to externalize metadata and adding ability to override the RTP display on game launch.
- Released new game title, All Aboard Panda Pennies, by cloning an existing game and extending it with responsive orientation support, full-screen background handling, and character group animations for compatible with its unique large-reel feature.
- Prepared entry into the German market by developing methodology for treating progressive pools as static jackpots, using scripts to manipulate progressive databases and implementing a customizable RTP display.
- Delivered 10+ staging site launches and 70+ game deployments by managing site-specific configurations across multiple jurisdictions.
- Shipped 15+ game re-releases, resolving 85+ issues to ensure production stability.

Projects

Kotoba Tag!

kotoba-tag.com

- Implemented an answer validation pipeline with ~95% accuracy by fine-tuning a SentenceTransformer model on 4M+ word pairs that were scraped, labeled and organized for cross-lingual definition similarity.
- Curated a 5,800+ word Japanese vocabulary bank with JLPT filtering, part-of-speech validation and syllable matching logic.
- Deployed a production application with XState, React Router, and a responsive shadcn/ui layout.
- Built a FastAPI backend on Render proxying HuggingFace inference and Jisho.org lookups, with cold-start warmup and retry handling.
- Enhanced gameplay with scoring multipliers, configurable game settings, and dictionary detail lookup.
- Maintained reliability with a Vitest suite covering syllable matching, definition validation and game flow.

Personal Website

nphach.github.io

- Built and deployed a Tamagotchi-inspired portfolio SPA with React, TypeScript, Vite, Tailwind CSS, and GitHub Actions CI/CD to GitHub Pages.
- Orchestrated an LCD zoom transition in Motion with viewport-aware geometry, game-inspired inventory UX, and Canvas pixel trails.

Image Analysis: Skin Detection

Spring 2024

- Improved a skin detection pipeline from ~73% to ~98% accuracy by applying color-space analysis and dynamic thresholding, refactoring the core algorithm, and adding async image processing to reduce runtime.
- Built a ground-truth skin-mask test dataset with Photoshop and implemented a Metrics class to benchmark baseline performance and evaluate detection accuracy against labeled annotations.

Education

Boston University, M.Sc in Software Engineering for AI

expected May 2028

University of Massachusetts Boston, B.Sc in Computer Science

May 2024